

The Constrained Continuum Emergence Framework: Ontology and Field Equations

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Abstract

The Constrained Continuum Emergence Framework (CCEF) describes gravity, gauge interactions, and matter as collective properties of a *single* continuum order-parameter field defined on a fixed background manifold. The order parameter takes values in $S^3 \cong SU(2)$; this target is not assumed but selected: requiring a compact, three-dimensional, simply-connected order-parameter space fixes it uniquely (Poincaré/Perelman), so $SU(2)$ and the Hopf fibration $S^3 \rightarrow S^2$ are outputs. The three Hopf projections of the single field furnish the three sectors: the base S^2 carries gravity, the $U(1)$ fibre is the emergent photon, and the field's own degree (π_3) is matter. The metric, the Einstein–Hilbert action, the composite graviton, the abelian gauge field, and localized *fermionic* solitons all appear as induced responses and topological excitations of this field. The observable universe corresponds to the disordered (symmetric) phase, in which the emergent abelian sector is in its Coulomb regime and the gauge boson is massless. This paper states the ontology and collects the defining equations: the order-parameter action, the induced gravitational sector, the composite graviton, the phase structure of the emergent abelian field, the (fermionic) matter sector and its cross-sector identities, the cosmological equations, the scale-invariant scalar spectrum, and the fixed parameter set.

1 Ontology

CCEF postulates a single fundamental entity: a continuum order-parameter field Φ defined on a fixed, non-dynamical four-manifold with a flat reference structure. Every observed structure is a property of Φ :

1. **Geometry is not fundamental.** The spacetime metric $g_{\mu\nu}$ is the source term conjugate to the field's stress tensor $T_{\mu\nu}$; it enters as a coupling, not as an independent degree of freedom.
2. **Gravity is an induced response.** Fluctuations of Φ generate an Einstein–Hilbert term for $g_{\mu\nu}$; Newton's constant is fixed by the field content and by the scale M at which the field's response becomes non-local.
3. **The graviton is composite.** The spin-two excitation is the transverse–traceless channel of the connected two-point function of $T_{\mu\nu}$. Conservation of $T_{\mu\nu}$ protects it at zero mass.
4. **Gauge fields are collective.** An abelian connection is built from the phase of Φ ; its dynamics follow from the four-derivative sector of the order-parameter action.
5. **Matter is topological.** Localized, particle-like states are finite-energy solitons of Φ , classified by the homotopy of its target space; their Fermi statistics are fixed by $\pi_4(S^3) = \mathbb{Z}_2$.
6. **Phases.** Φ admits a disordered (symmetric) phase and an ordered (broken) phase. The observable universe is the disordered phase. The two phases carry different low-energy content, in particular a massless versus a massive emergent vector boson.

The remainder of the paper makes these statements quantitative.

2 Field content

The target is derived, not assumed. Three requirements on the order-parameter manifold $\mathcal{M}$ fix it uniquely: $\mathcal{M}$ is *compact* (an order parameter is bounded), *three-dimensional* (a graviton needs an S^2 sector and a mass-viable soliton needs one further dimension), and *simply-connected* (no line defects, a clean topological sector). By the Poincaré conjecture (Perelman), the unique closed simply-connected three-manifold is S^3 . Hence $\mathcal{M} = S^3 \cong SU(2)$, and the Lie/chiral structure, the Hopf fibration, $\pi_3 = \mathbb{Z}$ (baryon number), $\pi_4 = \mathbb{Z}_2$ (fermion statistics), and parallelizability all follow as consequences rather than inputs.

The single field and its three projections. The order parameter is a single field $z = (z_1, z_2)$ valued in S^3 ,

$$z^\dagger z = 1, \quad z \sim e^{i\alpha} z, \quad (1)$$

so that $S^3/U(1) = S^2 = \mathbb{CP}^1$. The phase redundancy defines a composite abelian connection (the emergent photon),

$$a_\mu = -i z^\dagger \partial_\mu z, \quad f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu, \quad (2)$$

with gauge-covariant derivative $D_\mu z = (\partial_\mu - ia_\mu)z$. The Hopf *base* is the unit three-vector

$$\mathbf{n} = z^\dagger \boldsymbol{\sigma} z, \quad \mathbf{n} \cdot \mathbf{n} = 1, \quad (3)$$

with $\boldsymbol{\sigma}$ the Pauli matrices; it carries the *gravitational* sector. In this description the field strength of the fibre is

$$f_{\mu\nu} = \mathbf{n} \cdot (\partial_\mu \mathbf{n} \times \partial_\nu \mathbf{n}). \quad (4)$$

Thus a single S^3 field supplies all three sectors: the Hopf base $\mathbf{n} \in S^2$ is gravity, the Hopf fibre a_μ is the emergent photon, and the field's own degree (π_3) is matter. The gravitational sector is equivalently the $N = 2$ member of the $\mathbb{CP}^{N-1}$ family, for which $\mathbb{CP}^1 \cong S^2$.

3 Order-parameter action

The action is organized as an expansion in derivatives, with a two-derivative term, a marginal four-derivative sector suppressed by the non-locality scale M , and a potential:

$$S[\mathbf{n}] = \int d^4x [\mathcal{L}_2 + \mathcal{L}_4 + \mathcal{L}_V]. \quad (5)$$

Two-derivative term.

$$\mathcal{L}_2 = \frac{1}{2g^2} \partial_\mu \mathbf{n} \cdot \partial^\mu \mathbf{n}. \quad (6)$$

Four-derivative (marginal) sector. Define the independent quartic invariants

$$O_1 = (\partial_\mu \mathbf{n} \cdot \partial^\mu \mathbf{n})^2, \quad O_2 = (\partial_\mu \mathbf{n} \cdot \partial_\nu \mathbf{n})(\partial^\mu \mathbf{n} \cdot \partial^\nu \mathbf{n}), \quad O_4 = (\square \mathbf{n}) \cdot (\square \mathbf{n}). \quad (7)$$

The algebraic identity $f_{\mu\nu} f^{\mu\nu} = O_1 - O_2$ identifies the antisymmetric quartic invariant with the square of the emergent field strength. The marginal sector is

$$\mathcal{L}_4 = \frac{1}{M^2} [A_2(O_1 - O_2) + A_3 O_4] = \frac{A_2}{M^2} f_{\mu\nu} f^{\mu\nu} + \frac{A_3}{M^2} (\square \mathbf{n})^2. \quad (8)$$

The coefficient A_2 sets the coupling of the emergent abelian field, $e_B^{-2} = 4A_2$.

Potential. The potential fixes the phase structure and the mass gap:

$$\mathcal{L}_V = -\frac{1}{2}A_1 E_0^2(1 - n_3^2) - \frac{1}{4}A_4 E_0^2 C(\mathbf{n}), \quad (9)$$

with E_0 the intrinsic energy unit, A_1 the anisotropy coefficient, $C(\mathbf{n})$ the constraint term, and A_4 setting the gap $\Lambda = \sqrt{A_4} E_0$.

4 Induced gravitational sector

Integrating over order-parameter fluctuations generates a local effective action for the metric that couples to $T_{\mu\nu}$. Its curvature expansion is

$$\Gamma_{\text{grav}}[g] = \int d^4x \sqrt{-g} \left[-\frac{\Lambda_{\text{eff}}}{8\pi G} + \frac{R}{16\pi G} + \alpha_2 R^2 + \alpha_3 R_{\mu\nu} R^{\mu\nu} + \dots \right]. \quad (10)$$

The coefficient of the Einstein–Hilbert term is fixed by the field content and the non-locality scale M ,

$$\frac{1}{16\pi G} = \frac{N_{\text{eff}}}{6} \cdot \frac{M^2}{16\pi^2} \iff M_{\text{Pl}}^2 = \frac{N_{\text{eff}} M^2}{48\pi^2}, \quad (11)$$

with N_{eff} the effective number of contributing modes. The sign of the leading heat-kernel coefficient,

$$a_1 = \frac{N_{\text{eff}}}{6} R > 0, \quad (12)$$

fixes the positivity of the graviton kinetic term. Relation (11) places the non-locality scale at $M \simeq 1.8 \times 10^{19} \text{ GeV}$ for $N_{\text{eff}} \simeq 9$.

5 Composite graviton

The spin-two excitation is the transverse–traceless channel of the connected stress-tensor correlator,

$$h_{\mu\nu}(x) \longleftrightarrow \langle T_{\mu\nu}(x) T_{\rho\sigma}(0) \rangle_{\text{TT}} \equiv \Pi_{\mu\nu,\rho\sigma}^{\text{TT}}(x). \quad (13)$$

Conservation of the stress tensor,

$$\partial_\mu T^{\mu\nu} = 0 \implies p_\mu \Pi^{\mu\nu,\rho\sigma}(p) = 0, \quad (14)$$

is the Ward identity of the emergent diffeomorphism invariance. It fixes the transverse–traceless self-energy to a massless pole,

$$\Pi^{\text{TT}}(p) = \frac{Z_{\text{TT}}}{p^2} + (\text{regular}), \quad Z_{\text{TT}} > 0, \quad m_{\text{graviton}} = 0. \quad (15)$$

The pole persists in the disordered phase, where the spectrum carries a gap $m_{\text{gap}} \sim \Lambda$ and the pole sits below the two-particle continuum at $p^2 = 4m_{\text{gap}}^2$.

6 Phase structure and the emergent abelian field

The order parameter admits two phases distinguished by the condensate $\langle |\Phi| \rangle$:

Disordered (symmetric) phase. $\langle |\Phi| \rangle = 0$. The emergent abelian field is in the Coulomb regime; the composite connection a_μ propagates as a massless vector with coupling

$$e_B^2 = \frac{1}{4A_2} \approx 0.765. \quad (16)$$

This is the phase of the observable universe, and accounts for a massless gauge boson.

Ordered (broken) phase. $\langle |\Phi| \rangle = 1$. The abelian symmetry is realized in the Higgs regime; the composite vector acquires a mass

$$m_\rho = \sqrt{\frac{A_1}{2A_2}} E_0 \approx 152 \text{ MeV}. \quad (17)$$

The transition between the two phases is second order in the effective potential, with the condensate serving as the order parameter.

7 Matter sector

Localized states are finite-energy topological solitons of the single order-parameter field. The relevant homotopy groups of the S^3 target and its Hopf base classify the charges and statistics:

$$\pi_3(S^3) = \mathbb{Z} \quad \text{baryon number } B \text{ (Skyrme degree of } z), \quad (18)$$

$$\pi_4(S^3) = \mathbb{Z}_2 \quad \text{Fermi statistics of the baryon (Finkelstein–Rubinstein),} \quad (19)$$

$$\pi_2(S^2) = \mathbb{Z} \quad \text{magnetic monopole of the emergent photon,} \quad (20)$$

$$\pi_3(S^2) = \mathbb{Z} \quad \text{Hopf charge / magnetic helicity of the fibre.} \quad (21)$$

Crucially the baryon is a *fermion* through $\pi_4(S^3) = \mathbb{Z}_2$ alone, with no colour and no Wess–Zumino term. The n -sector energy functional supporting Hopf/skyrmion textures combines the two- and four-derivative terms of Section 3,

$$E[\mathbf{n}] = \int d^3x \left[\frac{1}{2g^2} \partial_i \mathbf{n} \cdot \partial_i \mathbf{n} + \frac{A_2}{M^2} f_{ij} f_{ij} \right], \quad (22)$$

whose stationary configurations at fixed charge define the soliton spectrum; the soliton mass scale is set by E_0 . The π_2 hedgehog is a stable, $\sim \text{GeV}$, magnetically charged boson, a distinctive prediction of the framework.

Cross-sector identity. Because gravity (base), electromagnetism (fibre), and matter (degree) are invariants of the *same* field z , their topological charges are locked. The sharpest instance is

$$\frac{1}{4\pi^2} \int a \wedge da = B^2, \quad (23)$$

that is, the magnetic helicity of the emergent photon equals the *square* of the baryon number. The left side is purely electromagnetic and the right purely baryonic; the identity follows from the Hopf composition law and has no counterpart in a theory with independent gauge and matter fields.

Leptons and the completion sector. The single S^3 field carries exactly one independent π_3 charge (baryon number); its Hopf charge is slaved to B^2 , so there is no independent lepton number and no light lepton among its projections. Light leptons are therefore not projections of the order parameter: they are chirally protected fundamental fermions of the ultraviolet completion (the $SU(2)_M$ sector at the scale M), whose lightness is a genuine chiral protection rather than a tuning. Matter is thus two-origin — baryons are emergent S^3 solitons, leptons are elementary completion-sector fermions. The neutrino mass scale then follows parameter-free from the derived seesaw scale $f = \sqrt{M\Lambda}$,

$$m_\nu \simeq \frac{\Lambda^2}{f} = \frac{\Lambda^{3/2}}{M^{1/2}} \approx 0.03 \text{ eV}, \quad (24)$$

in the observed range ($\sqrt{\Delta m_{\text{atm}}^2} \approx 0.05 \text{ eV}$).

8 Cosmological equations

The induced gravitational dynamics take the Einstein form,

$$G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu} = 8\pi G T_{\mu\nu}, \quad (25)$$

with $T_{\mu\nu}$ the stress tensor of the disordered-phase order parameter. For a homogeneous, isotropic background with metric $ds^2 = -dt^2 + a(t)^2 d\mathbf{x}^2$, the field equations reduce to

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho + \frac{\Lambda_{\text{eff}}}{3}, \quad \frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) + \frac{\Lambda_{\text{eff}}}{3}, \quad (26)$$

with ρ and p the energy density and pressure of the order-parameter field. The effective cosmological constant takes the value $\Lambda_{\text{eff}} \simeq 2.5 \times 10^{-47} \text{ GeV}^4$.

9 Scalar spectrum

The primordial scalar spectrum is set by the two-point function of the order-parameter stress tensor evaluated at the critical point of the disordered phase. Stress-tensor conservation (14) fixes $T_{\mu\nu}$ at scaling dimension $d = 3$, so the spectrum is scale-invariant and this result is *exact* — protected by the same conservation law, and reinforced topologically by reading the perturbation as the fluctuation of the conserved Hopf-charge density (a linking number is a scale-invariant, conformal observable):

$$n_s|_{\text{protected}} = 1. \quad (27)$$

The tilt therefore vanishes at the protected level; in particular n_s is *not* shifted to $1 - \eta$ by the order-parameter anomalous dimension η , which controls operator dimensions but not the spectral index. The amplitude is fixed separately, and parameter-free, by the normalization of the topological (Hopf-charge) seed at the $O(3)/\mathbb{CP}^1$ critical point,

$$A_s = \frac{25}{16} N_H^2 \left(\frac{3\Delta_\varepsilon}{8\pi}\right)^2 \approx 2.4 \times 10^{-9}, \quad \Delta_\varepsilon = 3 - \frac{1}{\nu} = 1.594, \quad (28)$$

within $\sim 14\%$ of the observed value, with no tuned parameters. The same conservation law (14) thus controls the flatness of the spectrum and the masslessness of the graviton.

10 Parameters

Table 1 lists the fixed parameters of the framework.

Summary

The framework specifies a single continuum order-parameter field valued in $S^3 \cong SU(2)$ — a target selected uniquely by compactness, three-dimensionality, and simple connectivity (Poincaré/Perelman) — with an action comprising a two-derivative term, a marginal four-derivative sector at the scale M , and a potential. The three Hopf projections of the one field furnish the three sectors. From this content the metric acquires an Einstein–Hilbert action with Newton constant fixed by (11); the spin-two excitation is the transverse–traceless stress-tensor channel, massless by (14); an emergent abelian field is massless in the disordered phase and massive in the ordered phase; matter states are topological solitons whose Fermi statistics come from $\pi_4(S^3) = \mathbb{Z}_2$, with the cross-sector identity $\frac{1}{4\pi^2} \int a \wedge da = B^2$; leptons are not projections of the field but chirally protected fermions of the completion sector, with a parameter-free neutrino scale $m_\nu \approx 0.03 \text{ eV}$; and the cosmological dynamics take the Einstein form with an

Symbol	Value	Role
E_0	122.5 MeV	intrinsic energy unit
$\Lambda = \sqrt{A_4} E_0$	231 MeV	mass gap
A_1	1	anisotropy coefficient ($= f^2/4$, graviton scale)
A_2	0.3268	marginal (abelian) coefficient
A_4	3.555	gap coefficient
$e_B^2 = 1/4A_2$	0.765	emergent abelian coupling
m_ρ	152 MeV	vector mass (ordered phase)
M	1.77×10^{19} GeV	non-locality / completion scale
M_{Pl}	2.435×10^{18} GeV	reduced Planck mass
N_{eff}	$\simeq 9$	effective mode number
Λ_{eff}	2.5×10^{-47} GeV ⁴	effective cosmological constant
$f = \sqrt{M\Lambda}$	2×10^9 GeV	seesaw / portal scale
$m_\nu \simeq \Lambda^{3/2}/M^{1/2}$	≈ 0.03 eV	neutrino mass (seesaw, parameter-free)
A_s	$\approx 2.4 \times 10^{-9}$	scalar amplitude (topological seed)
n_s	1	scalar index (protected, exact)
η	0.0375	$O(3)$ anomalous dimension (operator dim.; not the tilt)
$\ln(M/\Lambda)$	45.8	scale hierarchy (logarithmic)
$(\Lambda/M)^2$	1.7×10^{-40}	scale hierarchy (ratio)

Table 1: Fixed parameters of the framework.

effective cosmological constant. The scalar spectrum is exactly scale-invariant, $n_s = 1$ (protected by stress-tensor conservation), with a parameter-free amplitude $A_s \approx 2.4 \times 10^{-9}$ set by the topological seed.